DQH-BAN: DUAL-STREAM QUANTUM HEALTHCARE INTELLIGENCE (DQH-BAN-EXP-2026-V2)

STREAM A: Hybrid Quantum-Classical Ensemble

Network Flows -> Soft-Threshold & Fourier Coord
+ Multi-Scale 1D-CNN -> TCN -> BiGRU -> Attention
-> 4-Qubit PennyLane VQC + XGB + LGB + RF + LogReg
-> Out-Of-Fold Stacking (Regularized Meta-LR)
-> Validation-Calibrated Operating Threshold (FPR <= 1%)

STREAM B: Biomedical Intelligence

Physiological Signals (ECG, PPG, Resp)
-> Polarity-Aligned Bandpass & 15s HRV Context
-> 1D-CNN + BiGRU Encoders + SQI Masking
-> Gated Fusion -> AF / Non-AF Classification

HYPERLEDGER FABRIC & SECURITY ORCHESTRATION

Device Auth & Access Control | AES-256-GCM Encrypted Off-Chain Vault | SHA-256 Commitments